Binomial theorem (Theorem 5.1) for positive integral exponent (integral exponent means integer power). Now let us state the Binomial theorem for rational power.

Binomial Theorem for Rational Exponent

Theorem 5.4 (Binomial Theorem for Rational Exponent): For any rational number n,

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$$

for all real numbers x satisfying $|x| < 1$.

As the proof involves higher mathematical concepts, let us assume the theorem without proof and see some particular cases and solve some problems. In the theorem

1. By taking $-x$ in the place of x, we get

$$(1-x)^n = 1 - nx + \frac{n(n-1)}{2!}x^2 - \frac{n(n-1)(n-2)}{3!}x^3 + \cdots \qquad (|x| < 1)$$

2. By taking $-n$ in the place of n, we get

$$(1+x)^{-n} = 1 + (-n)x + \frac{(-n)(-n-1)}{2!}x^2 + \frac{(-n)(-n-1)(-n-2)}{3!}x^3 + \cdots$$

Hence

$$(1+x)^{-n} = 1 - nx + \frac{n(n+1)}{2!}x^2 - \frac{n(n+1)(n+2)}{3!}x^3 + \cdots \qquad (|x| < 1)$$

3. by taking $-x$ and $-n$ in the places of x and n, we get

$$(1-x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + \cdots \qquad (|x| < 1)$$

Even though we have explicitly mentioned that n is a rational number in the theorem, some of us may hesitate to use n in place of a general rational number. So we give the theorem using the representation $\frac{p}{q}(q \neq 0)$ for a rational number.

$$\begin{aligned}
(1+x)^{\frac{p}{q}} &= 1 + \frac{p}{q}x + \frac{\frac{p}{q}(\frac{p}{q}-1)}{2!}x^2 + \frac{\frac{p}{q}(\frac{p}{q}-1)(\frac{p}{q}-2)}{3!}x^3 + \cdots \\
&= 1 + \frac{p}{q}x + \frac{p(p-q)}{q^2 2!}x^2 + \frac{p(p-q)(p-2q)}{q^3 3!}x^3 + \cdots \qquad (|x| < 1) \\
(1-x)^{\frac{p}{q}} &= 1 - \frac{p}{q}x + \frac{p(p-q)}{q^2 2!}x^2 - \frac{p(p-q)(p-2q)}{q^3 3!}x^3 + \cdots \qquad (|x| < 1)
\end{aligned}$$

Though the theorem gives a formula to compute $(1+x)^n$, to solve numerical problems quickly we must remember and able to write certain expansions directly. Observation of the coefficient in each of such expansions will be very helpful in solving problems. Let us list some of them: (Try yourself!).

1. $(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots$
2. $(1-x)^{-1} = 1 + x + x^2 + x^3 + \cdots$
3. $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + 5x^4 + 6x^5 + \cdots$
4. $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + 5x^4 - 6x^5 + \cdots$

All the above expansions are valid only when $|x| < 1$.